

PRASHANT KUMAR MISRA

Senior AI Engineer · Generative AI & LLM Systems · Agentic AI · Backend Infrastructure

prashantmisraai@gmail.com | +91-9742516289 | linkedin.com/in/prashant-misra02 | github.com/prashantmisraai | India

PROFESSIONAL SUMMARY

Senior AI Engineer with 6+ years designing and shipping production-grade AI systems at enterprise scale. Deep expertise in LLM architectures, Retrieval-Augmented Generation (RAG), Agentic AI, and distributed backend engineering. Built and deployed Incident Commander Agent (LangGraph multi-agent, AWS), Enterprise RAG Agent (end-to-end, Dockerized), and a Demand Forecasting platform using classical ML. Track record of 35% retrieval accuracy gains, 40% hallucination reduction, and 30% latency optimization in production. Strong in system design, microservices, MLOps, and cloud-native deployment on AWS.

CORE TECHNICAL SKILLS

AI / LLM Large Language Models, RAG, Agentic AI, LangChain, LangGraph, LlamaIndex, HuggingFace, Prompt Engineering, LLM Evaluation

ML / Data Demand Forecasting, Time-Series (ARIMA, XGBoost, Prophet), NER, Embedding Models, FAISS, Pinecone, ChromaDB, Elasticsearch

Backend Python, Java, FastAPI, Spring Boot, RESTful APIs, Microservices, Distributed Systems, Event-Driven Architecture

Cloud / DevOps AWS (EC2, S3, Lambda, Bedrock), Docker, Kubernetes, CI/CD (GitHub Actions), Monitoring & Observability

Databases PostgreSQL, MongoDB, Vector Databases, Redis

PROFESSIONAL EXPERIENCE

Senior AI Engineer · Hitachi Digital Services Oct 2024 – Present · Bengaluru

Architected production LLM-powered enterprise knowledge and search platform serving thousands of internal users, enabling natural-language query over structured and unstructured enterprise data.

Designed end-to-end RAG pipelines with custom re-ranking and context compression, achieving

Engineered hallucination mitigation framework combining grounding checks and factual consistency scoring, reducing hallucination rate by

Cut P99 API latency by

Deployed AI microservices on AWS (ECS + Lambda) with auto-scaling, achieving 99.9% uptime across all production environments.

Built LLM evaluation harnesses (RAGAS-inspired) for automated benchmarking of response quality, faithfulness, and answer relevance.

Senior Software Engineer – AI · Quantiphi Analytics Jun 2023 – Oct 2024 · Mumbai

Productionized 3 GenAI solutions across document processing, customer service automation, and knowledge management, improving workflow automation efficiency by

Designed event-driven microservices architecture on AWS that scaled to 2× peak traffic with zero SLA degradation.

Improved LLM response relevance by

Led 4-engineer cross-functional team delivering GenAI features from architecture design through CI/CD deployment and production monitoring.

Introduced structured LLM evaluation pipelines (automated regression + human-in-the-loop) that cut production quality incidents by 35%.

Senior Software Engineer – AI & Backend · Marlabs Inc. Aug 2022 – Apr 2023 · Hyderabad

Built high-throughput ML data pipelines for enterprise healthcare clients, improving batch processing efficiency by

Integrated HL7/FHIR healthcare data standards across 5 enterprise systems, improving cross-system data reliability and interoperability by 20%.

Reduced production defects by 25% by establishing CI/CD best practices, code coverage gates, and automated regression suites.

Software Engineer · Capgemini Jul 2019 – Aug 2022 · Pune

Built scalable Java/Spring Boot backend systems across 10+ enterprise modules for global Fortune 500 clients.

Delivered ML-driven analytics integrations via robust REST APIs, enabling data-driven decision-making for client product teams.

Maintained 99.9% system uptime via proactive monitoring (CloudWatch, Splunk), automated alerting, and incident response runbooks.

Mentored 3 junior engineers; drove Agile delivery across distributed cross-geo squads.

TECHNICAL PROJECTS

Incident Commander Agent — Agentic AI / LangGraph / AWS ■ GitHub

Built a production-grade autonomous incident response agent using LangGraph multi-agent orchestration — agents triage, diagnose, notify, and remediate infrastructure incidents end-to-end without human intervention.

Architecture: modular agents/ layer (triage, root-cause, runbook), graph/ state machine (LangGraph), tools/ (AWS CloudWatch, PagerDuty, Slack integrations), Dockerized FastAPI API gateway.

Deployed on AWS with Docker + docker-compose; agent graph handles branching, retries, and graceful fallback to human escalation.

Stack: Python, LangGraph, LangChain, FastAPI, AWS SDK, Docker, Shell scripts for infra automation.

Enterprise RAG Agent — Production RAG / LLM / Docker ■ GitHub

End-to-end production RAG system with full test coverage (tests/), data ingestion pipeline, Dockerized deployment, and a clean app.py FastAPI endpoint.

Implements hybrid retrieval (dense + sparse), context compression, and re-ranking to maximize answer faithfulness on enterprise document corpora.

Designed for plug-and-play LLM backends (OpenAI, Bedrock, local models) via abstraction layer in src/.

Stack: Python, LangChain, FAISS/Pinecone, FastAPI, Docker, pytest.

Demand Forecasting Platform — Classical ML / Time-Series · Traditional ML

Built a multi-model demand forecasting system using ARIMA, XGBoost, LightGBM, and Facebook Prophet for retail/supply-chain use cases.

Implemented feature engineering pipeline: lag features, rolling statistics, seasonal decomposition, and external regressor integration.

Model selection framework benchmarks across RMSE, MAPE, and SMAPE; auto-selects best model per SKU/region segment.

Stack: Python, scikit-learn, XGBoost, Prophet, pandas, matplotlib, Jupyter.

Named Entity Recognition (NER) — End-to-End NLP ■ GitHub

End-to-end NER pipeline (data prep → training → inference) in a single Jupyter notebook; covers tokenization, IOB tagging, and evaluation with seqeval metrics.

Stack: Python, HuggingFace Transformers, spaCy, Jupyter.

EDUCATION

B.Tech in Computer Science Engineering · SRM University 2015 – 2019

CERTIFICATIONS & ROLE TARGETS

Pursuing: AWS Certified Machine Learning Specialty · Google Cloud Professional ML Engineer. Open to: Senior AI Engineer · Staff AI Engineer · LLM Engineer · GenAI Engineer · Applied AI Engineer · ML Engineer · AI Backend Engineer · Senior Software Engineer (AI/ML).